1. Justify whether the following prior-likelihood pairs are conjugate:

   (a) $p \sim \text{Beta}(a, b)$ and $X|p \sim \text{Geo}(p)$.

   (b) $\mu \sim U(-1, 1)$ and $X|\mu \sim N(\mu, 1)$.

   (c) $\beta \sim \text{Gamma}(a, b)$ and $X|\beta \sim \text{Gamma}(\alpha, \beta)$, where a, b, α are known constants.

2. Verify that the Gamma prior and Poisson likelihood are conjugate. Show that the resulting posterior mean can be written as a weighted average of the prior mean and a function of the data. Identify the function and give an interpretation of the result.

3. Compute the Jeffrey's prior for the exponential likelihood with rate $\lambda > 0$. Justify whether it is a proper or improper prior. Is the posterior proper?

4. Compute the posterior predictive density $p(x_{n+1}|X_1, \dots X_n)$ when $\mu \sim N(m, s^2)$ and $(X_1, \dots, X_n)|\mu \sim N(\mu, \sigma^2)$ are i.i.d., when m, s^2, σ^2 are known constants.

5. The `children` data set contains a bivariate, monthly time series spanning 1987–2002. The `exposed` column is the number of children aged 6–10 in Linz. The `y` column is the number of those children killed or seriously injured as pedestrians.

   (a) Perform an exploratory analysis of the data.

   (b) Propose a model for the probability of being killed or injured in a given month, assuming that the data are not temporally structured (as an aside, your exploratory analysis should show whether that assumption is reasonable). Compute a point estimate with 95% confidence interval for that probability.

6. Let y_t denote the number of children killed or injured in month t in the `children` data set. A simple model is to treat months as independent, with

$$y_t \sim \text{Pois}(\mu).$$

   (a) Perform a Bayesian analysis with prior $\mu \sim \text{Exp}(1)$ and determine a point estimate and 95% credible interval for μ.

   (b) Perform a sensitivity analysis, i.e. vary your prior parameter and see how results are affected.

7. Let $(x_1, \dots, x_n) \overset{i.i.d.}{\sim} N(\mu_0, \sigma^2)$ with known σ^2 and unknown μ_0. Show that

$$\phi_n(\mathbf{x}_{1:n}) := \mathbb{1}\{|\bar{\mathbf{x}} - \mu_0| > \varepsilon\}$$

   satisfies the requirements of the hypothesis test in the Bernstein–von Mises theorem for the unknown mean μ.

   Hint: Use the Law of Large Numbers.

8. Assume $(y_1, \dots, y_n) \overset{i.i.d.}{\sim} N(0, \sigma^2)$, with $\sigma^2 > 0$ unknown. Show that the inverse gamma distribution is conjugate for σ^2 and hence determine the posterior distribution of $\sigma^2|(y_1, \dots, y_n)$.

9. Assume $(y_1, \ldots, y_n) \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$, with $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ unknown. Show that the prior
$$p(\mu, \sigma^2) = p(\mu|\sigma^2)p(\sigma^2)$$
with
$$\begin{aligned} \mu|\sigma^2 &\sim N(m_0, M_0\sigma^2), \\ \sigma^2 &\sim \Gamma^{-1}(\nu_0/2, \sigma_0^2\nu_0/2), \end{aligned}$$
is conjugate for μ, σ^2. Hence determine their posterior distribution.

   Hint: Use the identities
$$\begin{aligned} \sum_{i=1}^n (y_i - \mu)^2 &= \sum_{i=1}^n (y_i - \bar{\mathbf{y}})^2 + n(\bar{\mathbf{y}} - \mu)^2, \\ \frac{m_0^2}{M_0} - \frac{m_n^2}{M_n} + n\bar{\mathbf{y}} &= \frac{(\bar{\mathbf{y}} - m_0)^2}{M_0 + 1/n}, \end{aligned}$$
   and make use of the M_n, m_n notation introduced in the lecture slides. Otherwise the computation is painful.

10. Verify that
$$\widehat{\mathrm{Cov}}(\widehat{\beta}_{\mathrm{LS}}) = \widehat{\sigma}^2(\mathbf{X}^T\mathbf{X})^{-1},$$
    where $\widehat{\beta}_{\mathrm{LS}}$ is the least squares estimator for the linear regression model
$$\mathbf{y} = \mathbf{X}\beta + \varepsilon, \;\; \varepsilon \sim N(0, \sigma^2 I_{n\times n}).$$